ACADEMIC PROJECT SHOWCASE

SkillBridge

A verified campus marketplace for student services

Software Engineering and Modeling
MIT World Peace University (MIT-WPU)
Computer Science and Engineering | Semester V | 2026–2027

Soham Patil · Ganesh Gaikwad
Aditya Sardesai · Tirth Gangani

Faculty Guide: Dr. Bhavana Tiple



INTRODUCTION

The campus service gap

Students have skills to offer, but campus buyers need a trusted way to discover, book, and complete services.

Discovery is fragmented

Buyers may rely on informal groups or referrals to find tutoring, design, technical, or creative help.

Trust needs evidence

The implementation adds student verification, seller role controls, order history, and transaction-backed reviews.

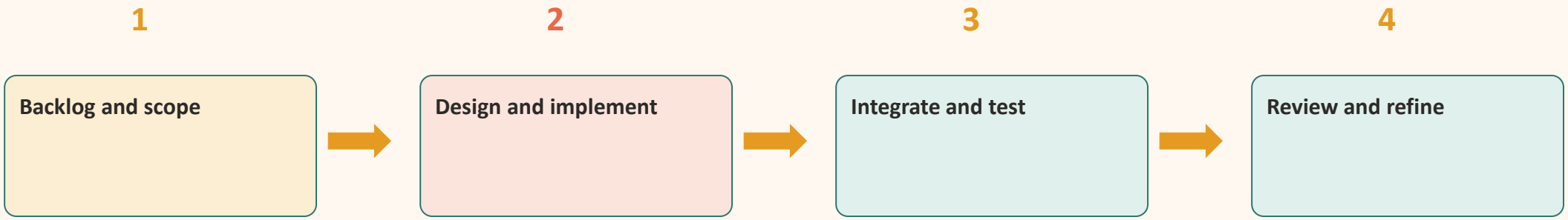
Service delivery needs structure

Listings, checkout, order milestones, private messaging, delivery review, and payment status are recorded in the system.

SkillBridge addresses these gaps through a campus-focused web marketplace with buyer, seller, and administrator workflows.

Development methodology

Agile Scrum supports short development cycles for a product with interconnected authentication, marketplace, payment, and governance features.



The deck describes Agile Scrum as the selected methodology. It does not infer sprint calendars, attendance, or retrospective results.

SRS

Scope and functional requirements

Area	Verified capability
Common	Register, authenticate, verify email, complete profile, switch buyer or seller role
Buyer	Explore published listings, use search and filters, create orders, pay in test mode, message seller, approve delivery, submit review
Seller	Create and manage listings, configure availability, receive and progress orders, submit deliverables, view earnings
Administrator	Review verification requests, moderate listings and reviews, view orders, users, metrics, and audit logs

SRS

Non-functional requirements and constraints

Security controls in code

Supabase authentication, protected routes, server-side role checks, RLS policies, participant-only order messages, and review eligibility checks.

Usability and compatibility

Responsive Next.js interface with browser form validation, route return parameters, clear empty states, notifications, and buyer/seller workspaces.

Operational constraints

Razorpay operates in test mode. AI search and transactional email have configuration-dependent fallbacks. No availability or response-time target is reported.

Claims above describe implementation controls or explicit constraints, not measured production service levels.

Current capabilities and project impact

Implemented capabilities

- Student account and verification workflow
- Buyer and seller modes within one profile
- Published service listings with availability
- Order lifecycle, escrow-style test payments, delivery and reviews
- Private order messaging and notifications
- Administrative moderation and audit visibility

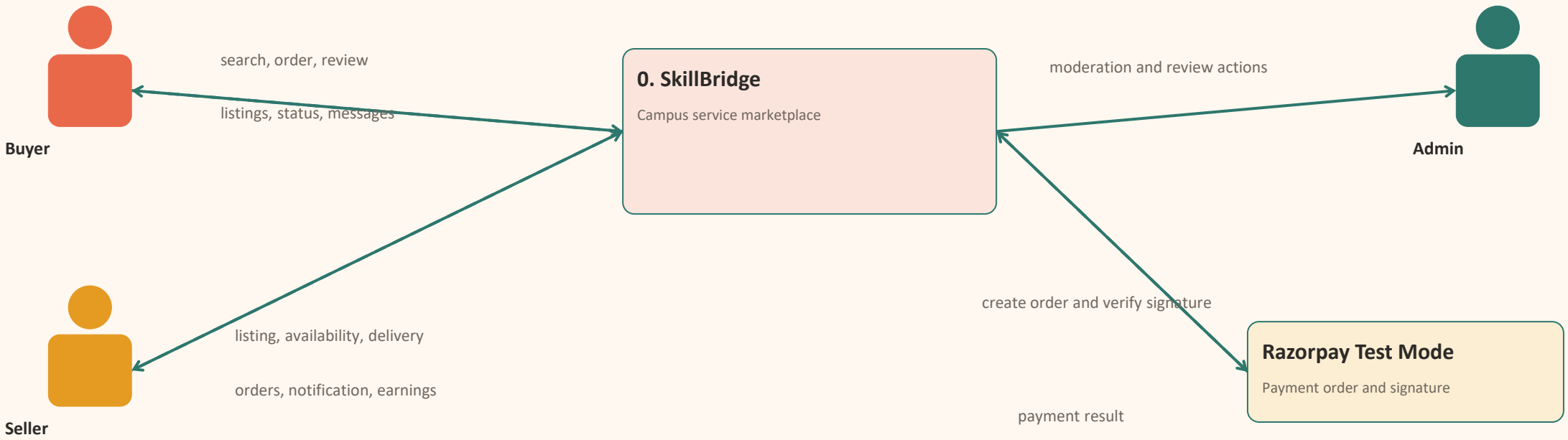
Future work

- Live payment onboarding beyond test mode
- Production email domain verification
- NVIDIA key configuration for AI-enhanced rankings
- Continued accessibility and performance testing

The product supports a structured peer-service exchange while preserving role boundaries and a traceable order record.

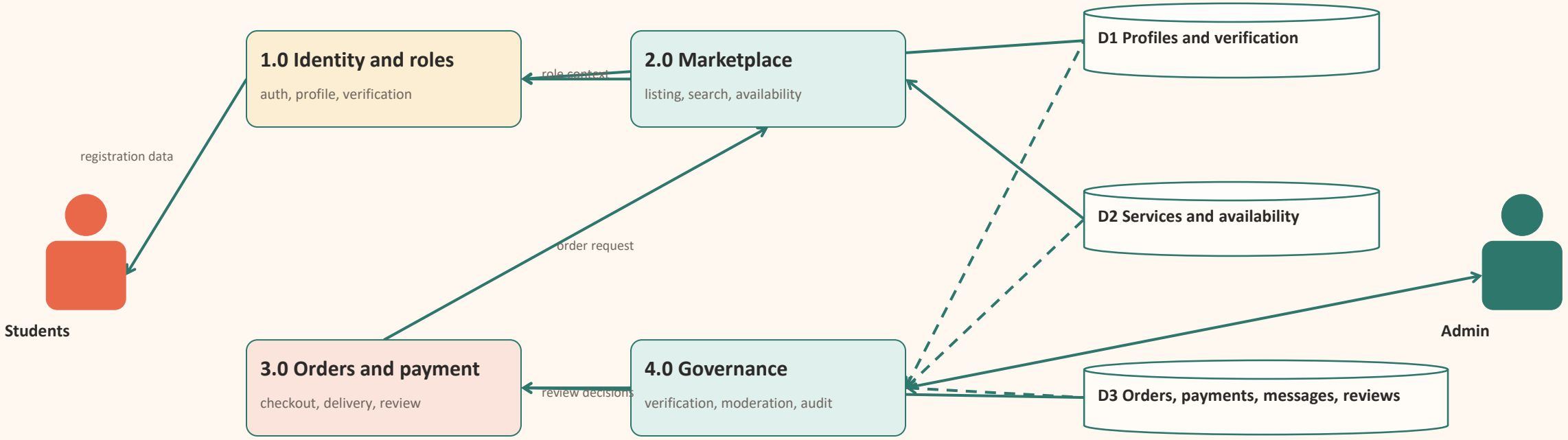
SYSTEM DESIGN

Data Flow Diagram: Level 0 context



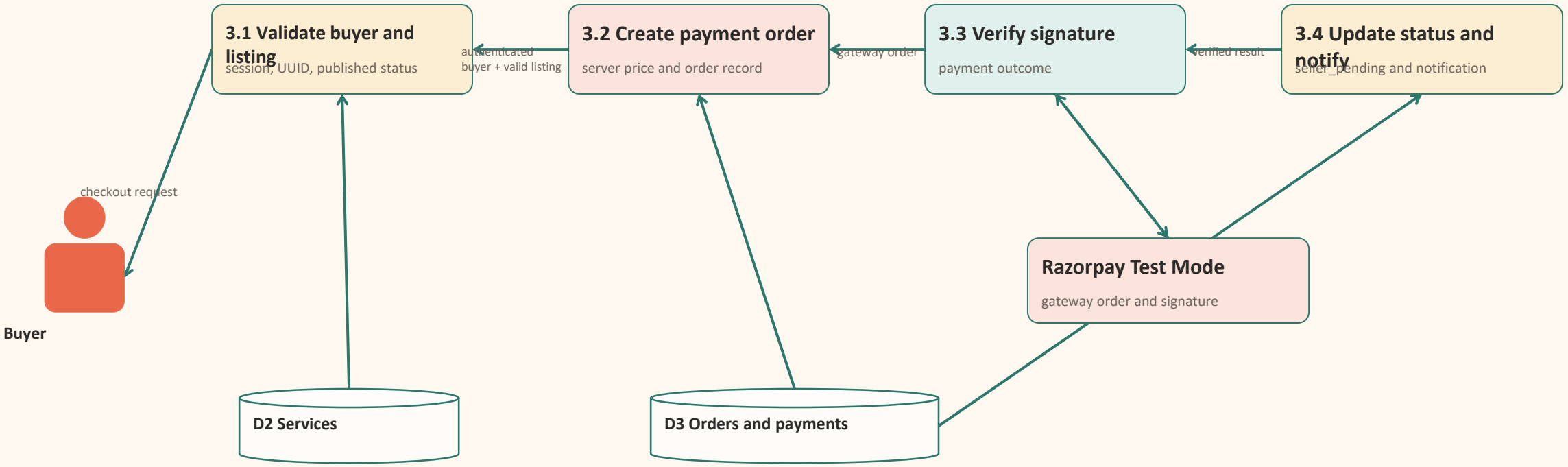
SYSTEM DESIGN

Data Flow Diagram: Level 1 decomposition



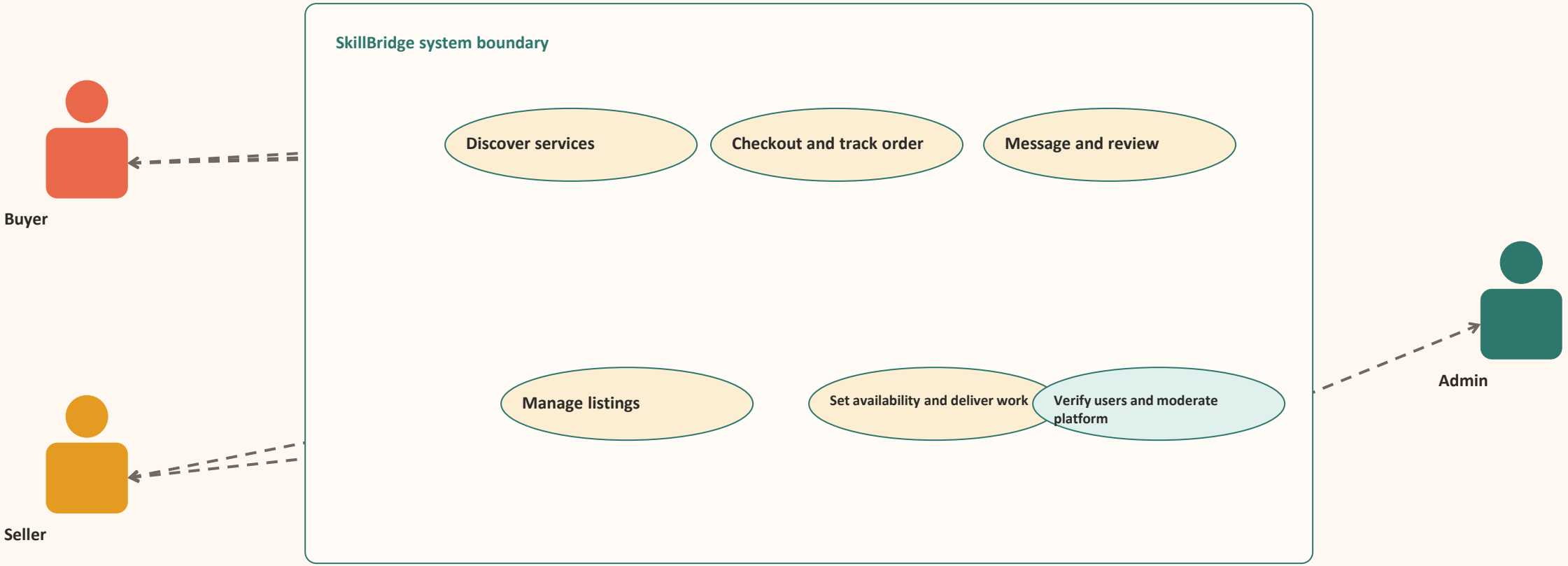
SYSTEM DESIGN

Data Flow Diagram: Level 2 checkout and payment verification



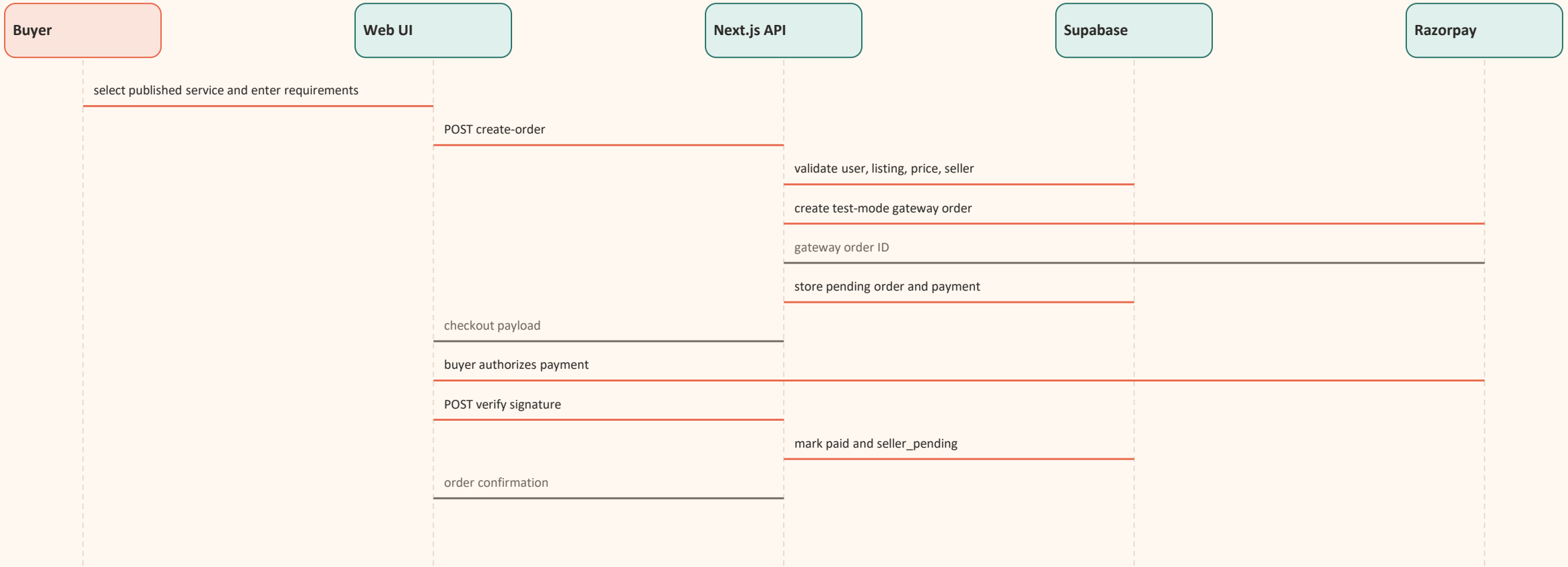
The API retrieves price and seller identity from the database; it does not accept a client-submitted amount.

Use Case Diagram



SYSTEM DESIGN

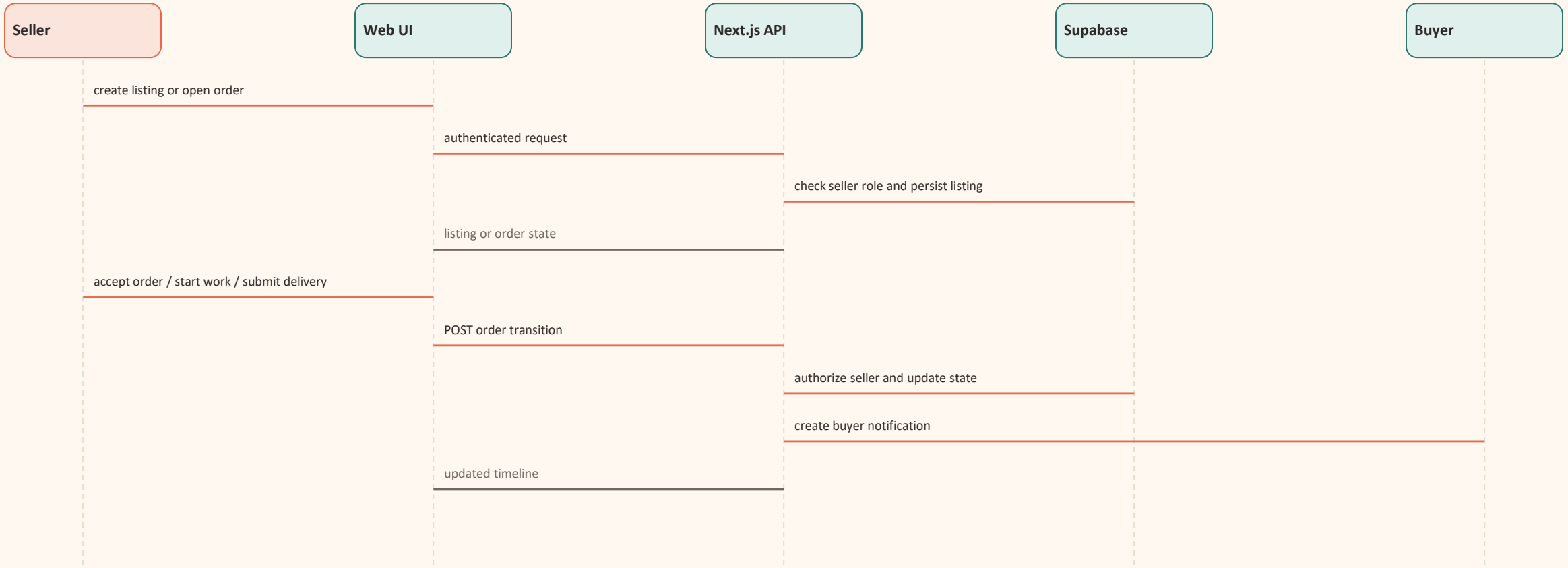
Sequence Diagram: Buyer checkout workflow



Alternate outcome: an invalid signature records a failed payment status and returns an error response.

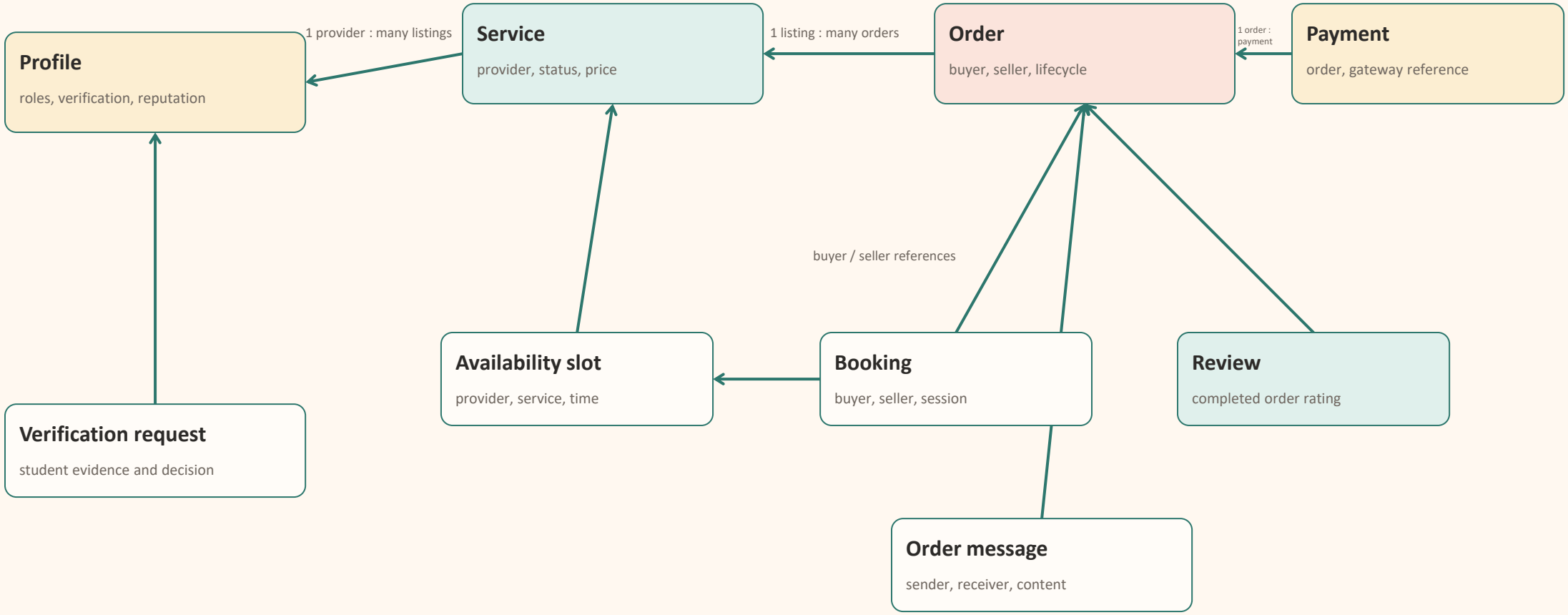
SYSTEM DESIGN

Sequence Diagram: Seller listing and delivery workflow

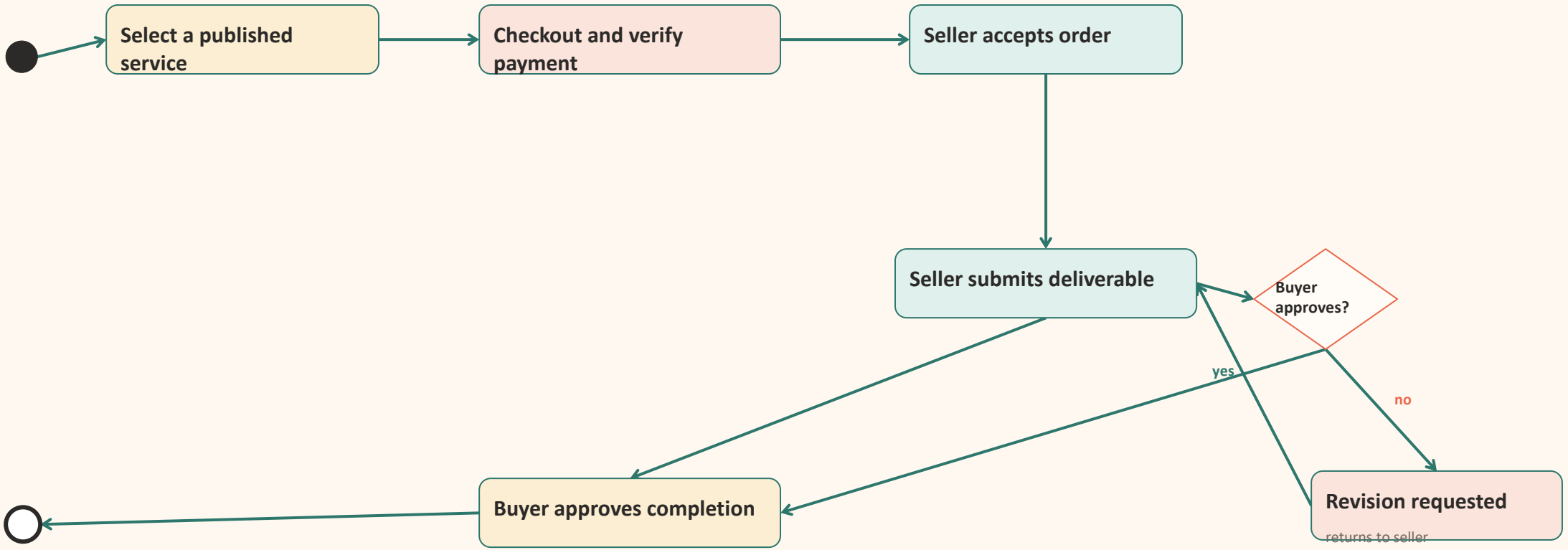


A seller cannot manage another provider's service, and a buyer approval triggers completion for submitted fixed-scope work.

Domain model



Activity Diagram: Fixed-scope order completion



Completion records the final order state and enables a transaction-backed buyer review.

ESTIMATION

Functional size estimation

Function Point estimate based on the implemented marketplace scope

Type	Count	Weight	UFP	Representative functions
EI	7	3	21	registration, verification, listing, availability, checkout, order action, review
EO	6	4	24	search results, dashboards, confirmations, notifications, rating output
EQ	4	3	12	catalog, profile, order, and message queries
ILF	8	7	56	profiles, services, availability, bookings, orders, payments, reviews, notifications
EIF	2	5	10	Razorpay and NVIDIA integration interfaces
Total			123	Unadjusted Function Points

UFP = 123. Assumed General System Characteristics total = 35. VAF = 0.65 + (0.01 × 35) = 1.00. Adjusted FP = 123 × 1.00 = 123.

EI: External Input | EO: External Output | EQ: External Inquiry | ILF: Internal Logical File | EIF: External Interface File

ESTIMATION

FP-to-KLOC conversion and Basic COCOMO

Functional size

123 adjusted FP

Conversion assumption

123 × 53 LOC/FP = 6,519 LOC = 6.52 KLOC

COCOMO mode

Basic COCOMO, organic mode

Measure	Formula	Estimated result
Effort	E = 2.4 × KLOC^1.05	17.3 person-months
Duration	T = 2.5 × E^0.38	7.4 months
Average staff	P = E / T	2.3 persons

This calculation estimates relative effort from functional size. It does not report actual team effort or project duration.

PLANNING

Approximate project plan and proposed responsibilities

Phase	Indicative focus	Status label
1. Scope	requirements, roles, data model, and user journeys	approximate
2. Build	frontend flows, API routes, Supabase schema, integrations	approximate
3. Integrate	orders, payments, messages, reviews, admin workflows	approximate
4. Validate	Selenium black-box evidence, documentation, presentation	approximate

Proposed allocation

Soham Patil — frontend and integration
Ganesh Gaikwad — backend and data model
Aditya Sardesai — testing and documentation
Tirth Gangani — UX, testing, and presentation support

To be confirmed by the team. This slide does not claim historical individual ownership.

TESTING

Testing strategy and environment

Method

Automated black-box functional testing. The test suite observes browser behavior, form constraints, toasts, routing, and visible state changes.

Tooling

Selenium WebDriver 4.51.0, pytest 9.1.1, Chrome 154 headless mode, and a local Next.js target backed by remote Supabase services.

Techniques

Equivalence partitioning, decision-table testing, state-transition testing, use-case testing, boundary-value analysis, and error guessing.

Execution evidence separates the initial full-suite baseline from a focused selected-test rerun.

TESTING

White-box testing scope and evidence limit

The available evidence supports browser-driven black-box testing. The presentation therefore does not claim measured white-box coverage.

What the repository provides

TypeScript application code, SQL migrations, test scripts, and Selenium tests. These support code inspection and traceability.

What is not claimed

No verified line, branch, path, or mutation coverage result. No synthetic coverage percentage is added.

Academic next step

Add a unit or integration test runner with coverage instrumentation, then report measured coverage separately from browser tests.

TESTING

Black-box testing: Equivalence Partitioning

ID	Input class	Expected observable result	Selected rerun
BB-EP-02	Valid email syntax + incorrect password	Credential error; remains on login page	PASS
BB-EP-04	Blank mandatory email	Browser blocks submission	PASS
BB-EP-07	Mismatched registration passwords	Mismatch message; remains on register page	PASS

The selected rerun verified representative valid and invalid input partitions. It does not replace the initial 39-test baseline.

TESTING

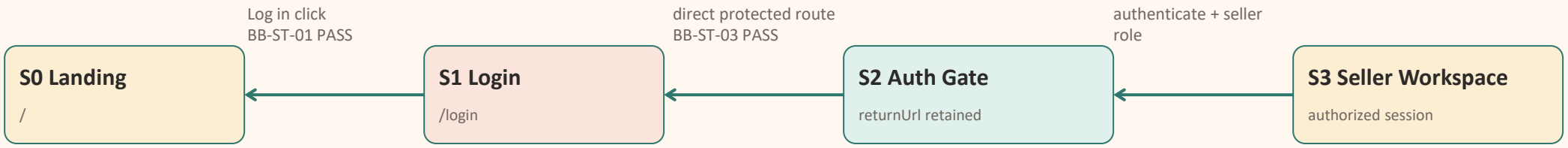
Black-box testing: Decision Table

Condition / action	Rule 1	Rule 2	Rule 3
Access target	/services/create	/services/create	/admin
Active session token	No	Yes	No
Required role	Seller	Seller	Administrator
Expected result	Redirect to login with returnUrl	Permit route evaluation	Redirect to login with returnUrl
Evidence	BB-DT-06 PASS	Role guard requires session and seller authorization	BB-EG-06 screenshot in suite

Decision-table cases model observable routing and role-entry conditions. Server API authorization adds a second boundary behind the UI route guard.

TESTING

Black-box testing: State transitions



Observed transition evidence confirms Landing → Login and unauthenticated Seller Route → Auth Gate. The last transition is shown as an implemented role-gated workflow rather than a selected-rerun claim.

TESTING

Black-box testing: Use cases and boundary values

ID	Technique	Scenario	Expected result	Selected rerun
BB-UC-04	Use case	Guest opens seller service-creation URL	Authentication gate protects seller workflow	PASS
BB-BVA-01	Boundary value	Registration password at 5 characters	Minimum length validation blocks submission	PASS
BB-BVA-02	Boundary value	Registration password at 6 characters	No minimum-length rejection	PASS

The BVA cases prove adjacent values around the configured six-character client-side threshold. They do not establish server-side password policy coverage.

TESTING

Test execution results

Initial full-suite baseline

39

Total tests

22

Passed

17

Failed / assertion discrepancies

Interpretation

The 17 baseline failures require triage. The supplied evidence identifies route-protection assumptions as one cause; failures must not be treated automatically as product defects.

Focused selected rerun

9 / 9 passed

Representative Selenium cases executed in 18.42 seconds

TESTING

Authentic Selenium execution evidence

Terminal: BB-EP-02 — Invalid Login Credentials

```
(base) student@skillbridge-workstation selenium_tests % python3 test_invalid_login.py
Target Endpoint: http://localhost:3000/login
Input Credentials: email=student.invalid@mitwpu.edu.in, password=[REDACTED]
Submitting authentication credentials...
Alert message: 'Invalid email or password. Please verify your credentials.'
User remained on login page: True (/login)
TEST RESULT: PASS
BLACK BOX TEST CASE [BB-EP-02]: PASS
```

Terminal: SkillBridge Selected Test Suite Pytest Summary

```
(base) student@skillbridge-workstation selenium_tests % python -m pytest tests/selected_test_suite.py -v
============================= test session starts ==============================
platform win32 -- Python 3.12.9, pytest-9.1.1, selenium-4.51.0, chrome-154.0.8037.98
rootdir: E:\MITWPU\Engineering\3rd Year\SEM\Project\project\skillbridge_blackbox_testing
collected 9 items

tests/test_equivalence_partitioning.py::BB_EP_02_invalid_password       PASSED [ 11%]
tests/test_equivalence_partitioning.py::BB_EP_04_empty_email_field      PASSED [ 22%]
tests/test_equivalence_partitioning.py::BB_EP_07_password_mismatch      PASSED [ 33%]
tests/test_state_transitions.py::BB_ST_01_landing_to_login_transition   PASSED [ 44%]
tests/test_decision_table.py::BB_DT_06_seller_unauth_guard              PASSED [ 55%]
tests/test_state_transitions.py::BB_ST_03_seller_gate_redirect          PASSED [ 66%]
tests/test_use_cases.py::BB_UC_04_seller_guard_workflow                 PASSED [ 77%]
tests/test_boundary_value_analysis.py::BB_BVA_02_password_exact_min     PASSED [ 88%]
tests/test_boundary_value_analysis.py::BB_BVA_01_password_below_min     PASSED [100%]

============================== 9 passed in 18.42s ==============================
Execution Verdict: 100% PASS RATE (9/9 Executed Cases)
```

Left: invalid credential handling | Right: selected 9-case Selenium rerun

CONCLUSION

Conclusion

SkillBridge delivers a verified campus marketplace with traceable buyer, seller, and administrator workflows.

- **Implemented** — Profiles and verification, service listings, availability, orders, test-mode payments, messaging, delivery, reviews, and moderation.
- **Testing evidence** — An initial Selenium baseline recorded 22 passes and 17 failures across 39 cases. A focused 9-case rerun passed.
- **Limitations** — Razorpay remains in test mode. AI search and email integrations depend on configuration. Full-suite failures need separate triage.
- **Next steps** — Resolve failed cases, collect code-coverage evidence, validate production configuration, and confirm individual project responsibilities.

Questions?

Thank you

SkillBridge
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